Limits and Continuity

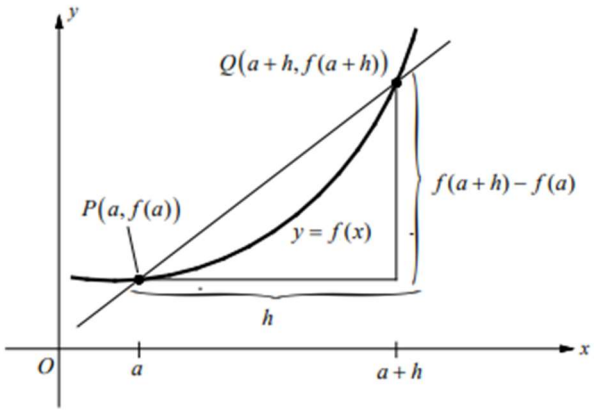
Rate of Change

Definition of Average Rate of Change

The expression

$$\frac{f(a+h)-f(a)}{h}$$

is called the **difference quotient** for f at a and represents the **average rate of change** of $y=f(x)$ from a to $a+h$.



Geometrically, the rate of change of f from a to $a+h$ is the **slope** of the secant line through the points $P(a,f(a))$ and $Q(a+h,f(a+h))$.

If $f(t)$ is the position function of a particle that is moving on a straight line, then in the time interval from $t=a$ to $t=a+h$, the change in position is $f(a+h)-f(a)$, and the average velocity of the particle over the time interval is

$$\textbf{average velocity} = \frac{\textbf{displacement}}{\textbf{time}} = \frac{f(a+h)-f(a)}{h}.$$

Exercises

1. The traffic flow at a particular intersection is modeled by the function f defined by $f(t)=25+6\cos($ for $0\le t\le 120$. What is the average rate of change of the traffic flow over the time interval $30\le t\le 40$

(A) 0.743 (B) 0.851 (C) 0.935 (D) 1.176

2. The rate of change of the altitude of a hot air balloon rising from the ground is given by $y(t)=t^3-3t^2+3t$ for $0\le t\le 10$. What is the average rate of change in altitude of the balloon over the time interval $0\le t\le 10$.

(A) 56 (B) 73 (C) 85 (D) 94

3.

t (sec)	0	10	20	30	40	50	60	70	80	90
$f(t)$ (ft/sec)	0	28	43	67	82	85	74	58	42	35

The table above shows the velocity of a car moving on a straight road. The car's velocity v is measured in feet per second.

(a) Find the average velocity of the car from $t = 60$ to $t = 90$.

(b) The instantaneous rate of change of f (See Ch. 2.1 for an explanation of instantaneous rate of change) with respect to x at $x = a$ can be approximated by finding the average rate of change of f near $x = a$. Approximate the instantaneous rate of change of f at $x = 40$ using two points, $x = 30$ and $x = 50$.

The Limit of a Function and One-Sided Limits

Definition of Limit

The statement

$$\lim_{x \to c} f(x) = L$$

means f approaches the **limit** L as x approaches c.

Which is read "the limit of $f(x)$, as x approaches c, equals L."

Basic Limits

1. If f is the constant function $f(x) = k$, then for any value of c,

$$\lim_{x \to c} f(x) = \lim_{x \to c} k = k.$$

2. If f is the polynomial function $f(x) = x^n$, then for any value of c,

$$\lim_{x \to c} f(x) = \lim_{x \to c} x^n = c^n.$$

One Sided Limits

The **right-hand limit** means that x approaches c from values greater than c.
This limit is denoted as

$$\lim_{x \to c^+} f(x) = L.$$

The **left-hand limit** means that x approaches c from values less than c.
This limit is denoted as

$$\lim_{x \to c^-} f(x) = L.$$

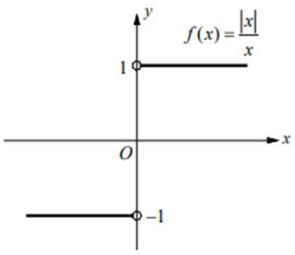
The Existence of a Limit

The limit of $f(x)$ as x approaches c is L if and only if

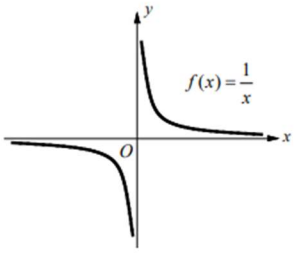
$$\lim_{x \to c^+} f(x) = L \quad \text{and} \quad \lim_{x \to c^-} f(x) = L.$$

Limits That Fail to Exists

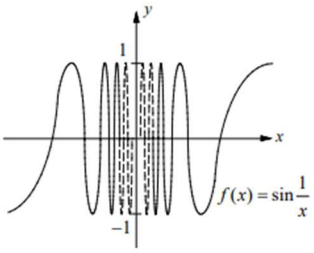
Some limits that fail to exists are illustrated below.



The left limit and the right limit is different. $\lim_{x\to 0}\frac{|x|}{x}=1$, if $x>0$ and $\lim_{x\to 0}\frac{|x|}{x}=-1$, if $x<0$.



As x approaches 0 from the right or the left, $f(x)$ increases or decreases without bound.



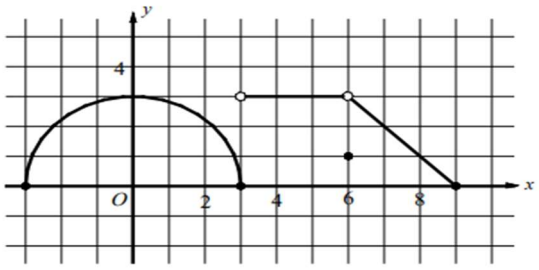
The values of $f(x)$ oscillate between -1 and 1 infinitely often as x approaches 0.

Exercises

1. The graph of the function f is shown in the figure below. Find the limit or value of the function at a given point.

(a) $\lim_{x\to 3^-} f(x)$ (b) $\lim_{x\to 3^+} f(x)$ (c) $\lim_{x\to 3} f(x)$

(d) $\lim_{x\to 6} f(x)$ (e) $f(3)$ (f) $f(6)$



Graph of $y=f(x)$

2. $\lim_{x\to\frac{\pi}{6}} \sec^2 x =$

(A) $\frac{3}{4}$ (B) $\frac{\sqrt{3}}{2}$ (C) $\frac{4}{3}$ (D) $\frac{2\sqrt{3}}{3}$

3. If $f(x)=\begin{cases} x^2+3, & x\neq 1 \\ 1, & x=1 \end{cases}$, then $\lim_{x\to 1} f(x) =$

(A) 1 (B) 2 (C) 3 (D) 4

4. $\lim_{x\to 1}\frac{|x-1|}{1-x} =$

(A) -2 (B) -1 (C) 1 (D) nonexistent

Limit Laws

Let c and k be real numbers and the limits $\lim_{x\to c} f(x)$ and $\lim_{x\to c} g(x)$ exist.

Then

1. $\lim_{x\to c}\left[f(x)\pm g(x)\right]=\lim_{x\to c} f(x)\pm\lim_{x\to c} g(x)$
2. $\lim_{x\to c}\left[f(x)\cdot g(x)\right]=\lim_{x\to c} f(x)\cdot\lim_{x\to c} g(x)$
3. $\lim_{x\to c}\left[k\,f(x)\right]=k\lim_{x\to c} f(x)$
4. $\lim_{x\to c}\dfrac{f(x)}{g(x)}=\dfrac{\lim_{x\to c} f(x)}{\lim_{x\to c} g(x)}$
5. $\lim_{x\to c}\left[f(x)\right]^n=\left[\lim_{x\to c} f(x)\right]^n$
6. $\lim_{x\to c}\sqrt[n]{f(x)}=\sqrt[n]{\lim_{x\to c} f(x)}$

Special Trigonometric Limits

1. $\lim_{x\to 0}\dfrac{\sin x}{x}=1$
2. $\lim_{x\to 0}\dfrac{1-\cos x}{x}=0$

Exercises

1. Find the limits.

(a) $\lim_{x\to 0}\dfrac{\sin 4x}{3x}$ (b) $\lim_{x\to 0}\dfrac{\tan x}{x}$ (c) $\lim_{x\to \pi/4}\dfrac{\tan x-1}{\sin x-\cos x}$

2. Find the limits.

(a) $\lim_{x\to 2}\dfrac{x^2+3x-10}{x-2}$ (b) $\lim_{x\to 1}\dfrac{\sqrt{x+3}-2}{x-1}$

3. Find the limits.

(a) $\lim_{x\to 0}\dfrac{\sin 4x}{3x}$ (b) $\lim_{x\to 0}\dfrac{\tan x}{x}$ (c) $\lim_{x\to \pi/4}\dfrac{\tan x-1}{\sin x-\cos x}$

4. $\lim_{\theta\to 0}\dfrac{\theta+\theta\cos\theta}{\sin\theta\cos\theta}=$

(A) $\dfrac{1}{4}$ (B) $\dfrac{1}{2}$ (C) 1 (D) 2

5. If $\lim_{x\to 0}\dfrac{\sqrt{2+ax}-\sqrt{2}}{x}=\sqrt{2}$ what is the value of a?

6. Find $\lim_{h\to 0}\dfrac{f(x+h)-f(x)}{h}$, if $f(x)=\sqrt{2x+1}$.

7. Find $\lim_{x\to 0}\dfrac{f(x)-g(x)}{\sqrt{g(x)+7}}$, if $\lim_{x\to 0} f(x)=2$ and $\lim_{x\to 0} g(x)=-3$.

8. Find $\lim_{x\to\sqrt{3}} g(x)$, if $\lim_{x\to\sqrt{3}}\dfrac{1}{x^2+g(x)}=\dfrac{1}{5}$.

9. $\lim_{x\to 0}\dfrac{\sqrt{4+x}-2}{x}=$

(A) $\dfrac{1}{8}$ (B) $\dfrac{1}{4}$ (C) $\dfrac{1}{2}$ (D) nonexistent

Definition of Continuity

A function f is **continuous at** c if the following three conditions are met.

1. $f(c)$ is defined
2. $\lim_{x\to c} f(x)$ exists
3. $\lim_{x\to c} f(x) = f(c)$

Intermediate Value Theorem

If f is continuous on the closed interval $[a,b]$ and k is any number between $f(a)$ and $f(b)$, then there is at least one number c in $[a,b]$ such that $f(c)=k$.

Specifically, if f is continuous on $[a,b]$ and $f(a)$ and $f(b)$ differ in sign, the Intermediate Value Theorem guarantees the existence of at least one zero of f in the closed interval $[a,b]$.

Exercises

1. For what values of a is $f(x)=\begin{cases} x^2, & x\le 1 \\ ax+2, & 1<x\le 3 \end{cases}$ continuous at $x=1$?

2. Let f be a function given by $f(x)=x^3-4x+2$. Use the Intermediate Value Theorem to show that there is a root of the equation on $[0,1]$.

3. Let f be a function defined by $f(x)=\begin{cases} \dfrac{x^2-a^2}{x-a}, & \text{if } x\neq a \\ 4, & \text{if } x=a \end{cases}$. If f is continuous for all real numbers x, what is the value of a?

(A) $\dfrac{1}{2}$ (B) 0 (C) 1 (D) 2

4. Let f be a continuous function on the closed interval $[-2,7]$. If $f(-2)=5$ and $f(7)=-3$, then the Intermediate Value Theorem guarantees that

(A) $f'(c)=0$ for at least one c between -2 and 7

(B) $f'(c)=0$ for at least one c between -3 and 5

(C) $f(c)=0$ for at least one c between -3 and 5

(D) $f(c)=0$ for at least one c between -2 and 7

5. Let g be a function defined by $g(x)=\begin{cases}\dfrac{\pi\sin x}{x}, & \text{if } x<0\\ a-bx, & \text{if } 0\le x<1\\ \arctan x, & \text{if } x\ge 1\end{cases}$.

If g is continuous for all real numbers x, what are the values of a and b?

6. Evaluate $\lim\limits_{a\to 0}\dfrac{-1+\sqrt{1+a}}{a}$.

7. What is the value of a, if $\lim\limits_{x\to 0}\dfrac{\sqrt{ax+9}-3}{x}=1$?

The statement $\lim\limits_{x\to c} f(x)=\infty$ means that $f(x)$ **approaches infinity as** x **approaches** c.
The statement $\lim\limits_{x\to c} f(x)=-\infty$ means that $f(x)$ **approaches negative infinity as** x **approaches** c.

Definition of Vertical Asymptote

If $f(x)$ approaches infinity (or negative infinity) as x approaches c from the right or the left, then the line $x=c$ is a vertical asymptote of the graph of f.

The graph of rational function given by $y=\dfrac{f(x)}{g(x)}$ has a **vertical asymptote** at $x=c$ if $g(c)=0$ and $f(c)\neq 0$.

Definition of Horizontal Asymptote

A line $y=b$ is a **horizontal asymptote** of the graph of a function $y=f(x)$ if either

$$\lim_{x\to\infty} f(x)=b \quad \text{or} \quad \lim_{x\to-\infty} f(x)=b.$$

The statement $\lim\limits_{x\to\infty} f(x)=L$ means that $f(x)$ has the **limit** L **as** x **approaches infinity**.

The statement $\lim\limits_{x\to-\infty} f(x)=L$ means that $f(x)$ has the **limit** L **as** x **approaches negative infinity**.

Exercises

1. Find all vertical asymptotes of the graph of each function.

(a) $f(x)=\dfrac{x}{x^2-1}$ (b) $f(x)=\dfrac{x^2-4x-5}{x^2-x-2}$

2. Find the limit.

(a) $\lim\limits_{x\to\infty}\dfrac{x^3-4x^2+7}{2x^3-3x-5}$ (b) $\lim\limits_{x\to\infty}\dfrac{\sqrt{4x^2+6x}}{3x-2}$

3. If $x=1$ is the vertical asymptote and $y=-3$ is the horizontal asymptote for the graph of the function f, which of the following could be the equation of the curve?

(A) $f(x)=\dfrac{-3x^2}{x-1}$

(B) $f(x)=\dfrac{-3(x-1)}{x+3}$

(C) $f(x)=\dfrac{-3(x^2-1)}{x-1}$

(D) $f(x)=\dfrac{-3(x^2-1)}{(x-1)^2}$

4. What are all horizontal asymptotes of the graph of $y=\dfrac{6+3e^x}{3-3e^x}$ in the xy- plane

(A) $y=-1$ only

(B) $y=2$ only

(C) $y=-1$ and $y=2$

(D) $y=0$ and $y=2$

5. Let $f(x)=\dfrac{3x-1}{x^3-8}$.

(a) Find the vertical asymptote(s) of f. Show the work that leads to your answer.

(b) Find the horizontal asymptote(s) of f. Show the work that leads to your answer.

6. Let $f(x)=\dfrac{\sin x}{x^2+2x}$.

(a) Find the vertical asymptote(s) of f. Show the work that leads to your answer.

(b) Find the horizontal asymptote(s) of f. Show the work that leads to your answer.

Definition of Derivatives

The **derivative** of a function f at x, denoted by $f'(x)$ is

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

if the limit exists.

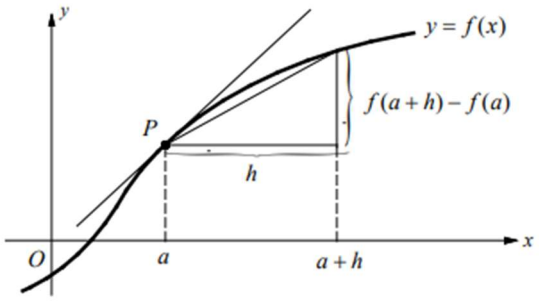
If we replace $x = a$ in the above equation, the derivative of a function f at a number a, denoted by $f'(a)$, is

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

If we write $x = a + h$, then $h = x - a$ and h approaches 0 if and only if x approaches a. Therefore, an equivalent way of stating the definition of derivative is

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

The process of finding the derivative of a function is called **differentiation**. In addition to $f'(x)$, which is read as "f prime of x," other notations such as $\frac{dy}{dx}$, y', and $\frac{d}{dx}[f(x)]$ are used to denote the derivative of $y = f(x)$.



Geometrically, the limit is the slope of the curve at P.

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

= slope of curve at P

= slope of tangent at P

Differentiability Implies Continuity

A function f is **differentiable** at a only if $f'(a)$ exists. If a function is differentiable at $x = a$, then f is continuous at $x = a$. Continuity, however, does not imply differentiability.

The derivative $f'(a)$ is the **instantaneous rate of change** of $f(x)$ with respect to x when $x = a$.

One-Sided Derivatives

The **left-hand derivative** of f at a is defined by

$$f'(a^-) = \lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h}, \text{ if the limit exist.}$$

The **right-hand derivative** of f at a is defined by

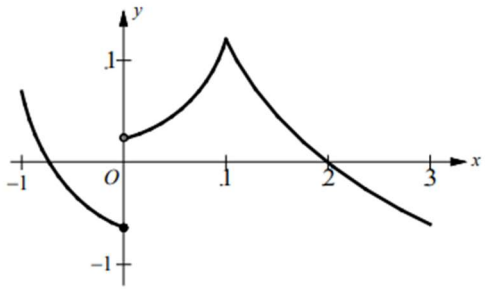
$$f'(a^+) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}, \text{ if the limit exist.}$$

$f'(a)$ exists if and only if these one sided derivatives exist and are equal.

Exercises

1. What is $\lim_{h\to 0}\dfrac{(3+h)^4-81}{h}$?

2. The graph of f is shown in the figure below. For what values of x, $-1<x<$ is f not differentiable?



3. $\lim_{h\to 0}\dfrac{(2+h)^5-32}{h}$ is

(A) $f'(5)$, where $f(x)=x^2$

(B) $f'(2)$, where $f(x)=x^5$

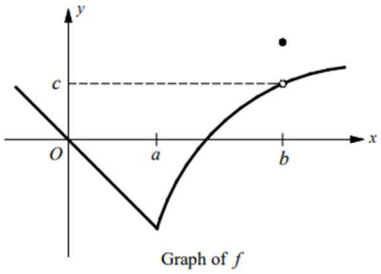
(C) $f'(5)$, where $f(x)=2^x$

(D) $f'(2)$, where $f(x)=2^x$

4. What is the instantaneous rate of change at $x=-1$ of the function $f(x)=-\sqrt[3]{x^2}$?

(A) $-\dfrac{2}{3}$ (B) $-\dfrac{1}{3}$ (C) $\dfrac{1}{3}$ (D) $\dfrac{2}{3}$

5.



Graph of f

The graph of a function f is shown in the figure above. Which of the following statements must be false?

(A) $f(x)$ is defined for $0\le x\le b$.

(B) $f(b)$ exists.

(C) $f'(b)$ exists.

(D) $\lim_{x\to a^-} f'(x)$ exists.

6. Let f be the function defined by $f(x)=\begin{cases} mx^2-2 & \text{if } x\le 1 \\ k\sqrt{x} & \text{if } x>1 \end{cases}$. If f is differentiable at $x=1$, what are the values of k and m?

7. If f is a differentiable function, then $f'(1)$ is given by which of the following?

I. $\lim_{h\to 0}\dfrac{f(1+h)-f(1)}{h}$

II. $\lim_{x\to 1}\dfrac{f(x)-f(1)}{x-1}$

III. $\lim_{x\to 0}\dfrac{f(x+h)-f(x)}{h}$

(A) I only (B) II only (C) I and II only (D) I and III only

8. Let f be a function that is differentiable throughout its domain and that has the following properties.

(1) $f(x+y)=f(x)+x^3y-xy^3-f(y)$

(2) $\lim_{x\to 0}\dfrac{f(x)}{x}=1$

Use the definition of the derivative to show that $f'(x)=x^3-1$.

9. Let f be the function defined by

$$f(x)=\begin{cases} x+2 & \text{for } x\le 0 \\ \dfrac{1}{2}(x+2)^2 & \text{for } x>0. \end{cases}$$

(a) Find the left-hand derivative of f at $x=0$.

(b) Find the right-hand derivative of f at $x=0$.

(c) Is the function f differentiable at $x=0$? Explain why or why not.

(d) Suppose the function g is defined by

$$g(x)=\begin{cases} x+2 & \text{for } x\le 0 \\ a(x+b)^2 & \text{for } x>0, \end{cases}$$

where a and b are constants. If g is differentiable at $x=0$, what are the values of a and b?

The Product Rule

If f and g are both differentiable, then $\dfrac{d}{dx}\left[f(x)g(x)\right]=f(x)g'(x)+g(x)f'(x)$.

The Quotient Rule

If f and g are both differentiable, then $\dfrac{d}{dx}\left[\dfrac{f(x)}{g(x)}\right]=\dfrac{g(x)f'(x)-f(x)g'(x)}{\left[g(x)\right]^2}$.

First derivative:	y',	$f'(x)$,	$\dfrac{dy}{dx}$
Second derivative:	y'',	$f''(x)$,	$\dfrac{d^2y}{dx^2}$
Third derivative:	y''',	$f'''(x)$,	$\dfrac{d^3y}{dx^3}$
Fourth derivative:	$y^{(4)}$,	$f^{(4)}(x)$,	$\dfrac{d^4y}{dx^4}$
$\vdots$			
***n*th derivative:**	$y^{(n)}$,	$f^{(n)}(x)$,	$\dfrac{d^ny}{dx^n}$

Exercises

1. Differentiate the function $f(x) = (x^3 - 7)(x^2 - 4x)$.

2. If $f(x) = \frac{1}{6}x^3 + 24\sqrt{x}$, find $f'(x)$, $f''(x)$, $f'''(x)$, and $f'''(9)$.

3. If $f(x) = (x^3 - 2x + 5)(x^{-2} + x^{-1})$, then $f'(1) =$

(A) -10 (B) -6 (C) $-\frac{9}{2}$ (D) $\frac{7}{2}$

4. If $g(2) = 3$ and $g'(2) = -1$, what is the value of $\frac{d}{dx}\left(\frac{g(x)}{x^2}\right)$ at $x = 2$?

(A) -3 (B) -1 (C) 0 (D) 2

5. If $y = 4\sqrt{x} - 16\sqrt[4]{x}$, then $y'' =$

(A) $\sqrt[4]{x} - 3$ (B) $-3\sqrt{x} + 3$ (C) $\frac{-\sqrt[4]{x} + 3}{x\sqrt[4]{x^3}}$ (D) $\frac{\sqrt{x} - 3}{x\sqrt[4]{x}}$

6. Let $h(x) = x \cdot f(x) \cdot g(x)$. Find $h'(1)$, if $f(1) = -2$, $g(1) = 3$, $f'(1) = 1$, and $g'(1) = \frac{1}{2}$.

7. Let $g(x) = \frac{x}{\sqrt{x} - 1}$. Find $g''(4)$.